28/10/25 → X: RV & its distribution

"unknown" in most modelling scenarios.

Usually, μ_X, σ_X^2 & second-order statistics

$Pr(X > t)$: bound using μ_X, σ_X^2

Markov equality: $\boxed{X \geq 0}$

Then, $P(X > t) \leq \frac{E[X]}{t}$

Proof:

$$X = X\,\mathbb{I}(X \leq t) + X\,\mathbb{I}(X > t)$$

$$E[X] = \underbrace{E[X\,\mathbb{I}(X \leq t)]}_{\geq 0} + E[X\,\mathbb{I}(X > t)]$$

$$E[X] \geq E[X\,\mathbb{I}(X > t)]$$

$$\geq E[t\,\mathbb{I}(X > t)]$$

$$= t\,P(X > t)$$

$\square$

$X \sim$ pdf f_X

$$P(X > t) = \int_t^\infty f_X(x)\,dx$$

$$E[X] = \int_0^\infty x f_X(x)\,dx = \int_0^t \underbrace{x f_X(x)\,dx}_{\geq 0} + \int_t^\infty x f_X(x)\,dx$$

$$\geq t\,P(X > t)$$

Chebyshev equality: $P((X - \mu_X)^2 > t^2) \leq \frac{\sigma_X^2}{t^2}$

X: RV with mean μ_X, var σ_X^2

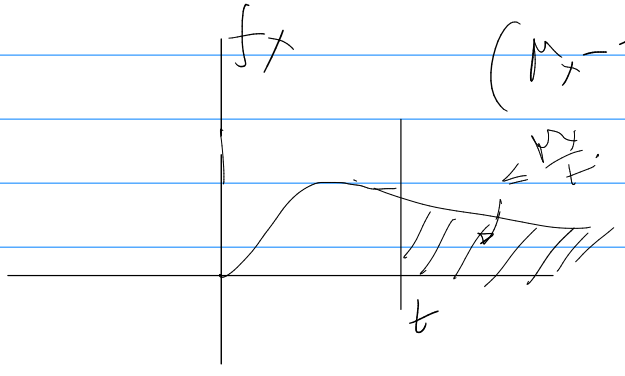
$(t > 0)$

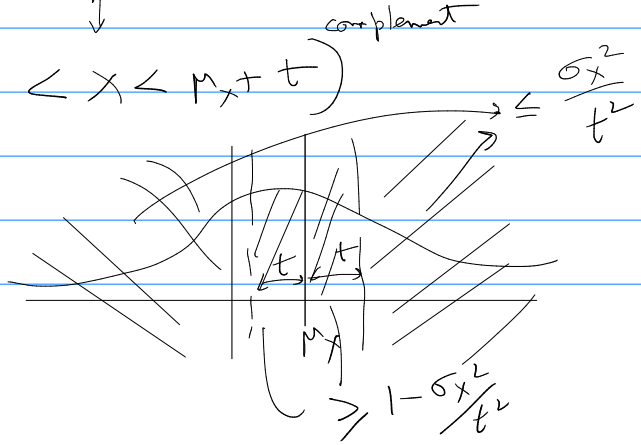
$\Updownarrow$

$|X - \mu_X| > t$

$\Updownarrow$

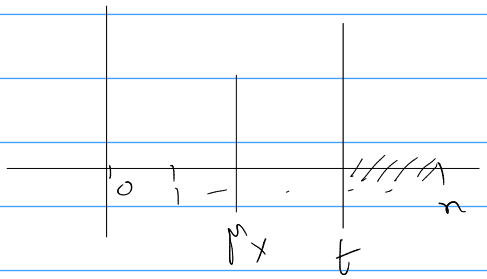
$(\mu_X - t < X < \mu_X + t)$ complement





Ex: 0 $X \sim Binom(n, p)$ $\mu_X = np$

$\sigma_X^2 = np(1-p)$



$$P(X \geq t) = \sum_{k=\lceil t \rceil}^{n} \binom{n}{k} p^k (1-p)^{n-k}$$

Markov: $P(X > t) \leq \frac{np}{t}$

Cheby: $P(|X - np| > t_0) \leq \frac{np(1-p)}{t_0^2}$

$P(X < np - t_0) + P(X > np + t_0)$

$\Rightarrow P(X > np + t_0) \leq \frac{np(1-p)}{t_0^2}$

Ex: $n = 1000$, $p = 0.01$

$P(X > 500) \leq \frac{10}{500}$ Markov

$P(X > 500) \leq \frac{9.9}{490^2}$

$X \sim N(0,1)$

$P(X > 10) \leq \frac{1}{100}$ cheby

$X = U_1 + U_2 + \cdots + U_n$ $\quad U_i \in Uniform[0,1]$ indep.

$\mu_X = n/2$ $\quad \sigma_X^2 = n/12$

$P(X > \frac{3n}{4}) \overset{Markov}{\leq} 2/3$

$\overset{cheby}{\leq} \frac{n/12}{(n/4)^2} = \frac{4}{3n}$

$X = V_1 + V_2 + \cdots + V_n$ $\quad V_i \sim \text{Exp}(\lambda)$ indep $\quad \lambda e^{-\lambda x}$

$\mu_X = \frac{n}{\lambda}, \sigma_X^2 = \frac{n}{\lambda^2}$

$P(X > cn) \overset{\text{Markov}}{\leq} \frac{1}{\lambda c}$

$(c > 1/\lambda)$

$\overset{\text{Chebyshev}}{\leq} \frac{n/\lambda^2}{n(c-\frac{1}{\lambda})^2}$

Chernoff bound $\quad X$: RV, $e^{tX} > 0$

$P(X > s) = P(e^{tX} > e^{ts}) \leq E[e^{tX}] \cdot e^{-ts}$

$(t > 0)$

$(t > 0) \quad tX \updownarrow st$

$e^{tX} > e^{st}$

$P(X > s) \leq \inf_{t>0} E[e^{tX}] e^{-ts}$

MGF: $M_X(t)$

[Hoeffding's Lemma: $X \in [a,b]$ $\quad M_{X-E[X]}(t) \leq e^{\frac{-t^2}{8(b-a)^2}}$]

Ex: $X \sim N(0,1)$

$P(X > 10) \leq e^{-50}$

$P(X > s) \leq \inf_{t>0} e^{t^2/2} \cdot e^{-st}$

$= e^{-s^2/2}$